Design and Analysis of Two-Layer Coding Scheme for DNA-Based Data Storage

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***Abstract*— DNA-based data storage has emerged as a compelling alternative to traditional media due to its ultra-high information density and long-term stability. However, the high read cost caused by the error-prone synthesis, storage, and sequencing processes remains a major bottleneck for practical deployment. To address this challenge, this paper proposes a read-cost-efficient coding framework that enhances reliability without increasing total redundancy. First, a novel two-layer intra-oligo coding scheme based on Bose–Chaudhuri–Hocquenghem (BCH) codes is presented, where index bits and data bits are respectively protected to mitigate base-level errors. Second, a semi-analytical optimization method based on the normal approximation of the finite-length coding rate is developed to allocate redundancy between index and data bits optimally under a fixed total code rate. The inter-oligo protection is further achieved through low-density parity-check (LDPC) codes to combat sequence-level errors. We then present extensive analytical and numerical results to show the effectiveness of the proposed analysis. Finally, we present numerical results to demonstrate that the concatenated code based on the optimized two-layer coding scheme significantly outperforms the concatenated code based on the single-layer coding scheme in terms of frame error rate (FER) under the same sequencing depth and total redundancy. These results underscore the advantages of the two-layer coding scheme and the optimization method for DNA-based data storage systems.**

***Index Terms*— DNA-based data storage, two-layer coding scheme, redundancy allocation, normal approximation.**

I. Introduction

IN THE age of data proliferation, traditional data storage technologies are insufficient to meet the exponential demands for storage capacity. DeoxyriboNucleic Acid (DNA), the natural genetic information storage medium, has an extremely high information density and stability, making it the ideal storage medium for big data [1], [2], [3], [4]. The longevity and information density of DNA molecules are determined by their unique structure. The high stability of DNA-based data storage results from the double-stranded structure formed by phosphate-linked nucleotides. The high information density of DNA-based data storage stems from the rich choices of nucleotides in each base pair position [5], [6]. It had been shown theoretically that only 1.0 kg of DNA can store all the digital information in the world until 2016 [7]. These merits of DNA-based data storage make it an attractive alternative to conventional long-term storage systems, i.e., hard-disk drives and magnetic tapes [7], especially when the speeds of synthesis and sequencing technologies in molecular biology are enhanced [8].

DNA-based data storage is well suited for archival applications involving large-scale data, such as historical records, scientific datasets, and cultural heritage, due to its high density and long-term durability [9], [10]. It is also advantageous in extreme environments where traditional electronic storage media are imperfect, including space missions [11]. Moreover, DNA storage offers the potential for secure data storage and steganography, exploiting the complexity of DNA sequences to conceal information [12].

Despite its potential, DNA-based data storage is still in its early stages and faces several challenges before practical deployment. First, the storage process involves multiple stages—binary-to-DNA encoding, synthesis, decay, polymerase chain reaction (PCR), sampling, sequencing, and DNA-to-binary decoding—each of which can introduce bit errors. Second, the original order of the data is not inherently preserved in the oligo pool, complicating accurate sequence retrieval. Third, the high cost of DNA synthesis and sequencing remains a major barrier, with costs closely linked to the amount of redundancy used for error correction. To address these challenges, effective error-correcting codes and indexing methods that improve the reliability of DNA-based data storage systems without excessive redundancy are essential.

A. State-of-the-Art Review

There are two types of methods to guarantee the fidelity of the data in DNA-based data storage. The first type of method is adding physical copies during the encoding process. The second type of method is introducing logical redundancy during the process of binary-to-DNA encoding. In the seminal work of Goldman et al. [13], the DNA molecules were synthesized with 4-fold redundancy. In Church et al. [14] early study, a sequencing depth of 3000× was considered, which greatly increased the physical redundancy. However, relying solely on physical redundancy to enhance the reliability of

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DNA-based data storage is impractical. This is due to the fact that the costs of DNA synthesis and sequencing increase linearly with the physical redundancy [15]. A more efficient approach to enhance the reliability in DNA-based data storage is to introduce logical redundancy into the sequence, akin to the error-correcting codes used in traditional communication systems.

In 2014, Yim et al. [16] introduced the low-density parity-check (LDPC) codes to DNA-based data storage. In their study, only substitution errors were considered, and the insertion and deletion (Indels) errors were assumed to be combated by majority voting, which requires a relatively high sequencing depth, i.e., from 104× to 104938×. Grass et al. [17] introduced Reed-Solomon (RS) code to DNA-based data storage. To effectively combat both base-level and sequence-level errors, they expanded the RS code to two-dimensional encoding, in which both inter-oligo and intra-oligo codes were employed.

In [18], the DNA fountain code [19] was advocated to combat the errors due to sequence loss during data storage. The DNA fountain code achieved a performance close to the Shannon capacity and offered significant resilience to data corruption. However, the major limitation of the DNA fountain code lies in the potential decoding failure when encountering stopping sets of Luby transform codes [20]. The DNA storage channel can also be modeled by the insertion/deletion/substitution (IDS) channel proposed by Davey and Mackay [21]. In 2021, Chen et al. [22] proposed to use watermark code in DNA-based data storage. In DNA watermark code, a pseudo-random watermark sequence was superimposed onto the information sequence before synthesis. When retrieving the data, a hidden Markov model (HMM) with the forward and backward algorithm was used to obtain the soft information of the original sequence. In 2022, Ping et al. [20] proposed a transcoding algorithm named the yin-yang codec, which can generate DNA sequences that are highly compatible with synthesis and sequencing technologies, thereby reducing the error rate of the retrieved messages. In 2024, Zhang and Zhou [23] proposed a new error correction method that integrated minimum Hamming distance, multiple sequence alignment (MSA), and an encoding scheme based on the ASCII-DNA encoding table. More recently, machine learning techniques have been integrated into the error-correcting processes of DNA-based data storage systems [24], [25], [26]. Using both inter-oligo and intra-oligo codes (2-D coding scheme) to encode messages is an efficient way to combat both sequence-level errors and base-level errors in DNA-based data storage [17], [27], [28], [29], [30]. In 2016, Blawat et al. [27] developed an efficient and robust forward error-correcting (FEC) scheme for intra-oligo encoding. In 2020, Antkowiak et al. [28] proposed a two-dimensional RS coding scheme. They utilized Reed-Solomon codes for both intra- and inter-oligo coding. They also utilized pseudo-randomization to avoid homopolymers and achieve the so-called GC balance.

In 2-D coding schemes, an index is inserted into each sequence before the intra-oligo coding to identify them when retrieving the data stored in DNA [31], [32], [33]. However, existing coding schemes encode the index bits and the data bits by a single error-correcting code. This type of coding scheme is referred to as the single-layer coding scheme in this paper. It is well-known that the importance of index bits is higher than that of the data bits due to the following voting procedure for combating insertion and deletion errors. To improve the reliability of index bits, the cyclic-redundancy-check was added to the index bits for error checking [16]. An enhanced single-layer coding scheme was then presented in [27] by further adding an FEC for the index bits. This scheme couples the encoding of index bits and data bits in intra-oligo coding. To the best of our knowledge, there are no well-established methods to optimize the allocation of the redundancies for index bits and data bits when a total amount of redundancy is given.

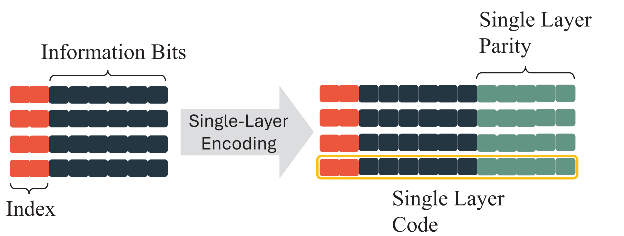
B. Contributions

Motivated by the observations above, this paper proposes a new intra-oligo coding scheme in which the index bits and the data bits are protected separately. Designed to mitigate base-level errors in DNA-based data storage, the resulting scheme is referred to as the two-layer intra-oligo coding scheme. Fig. 1 illustrates the conventional single-layer scheme alongside the proposed two-layer scheme. For a given total redundancy, we introduce a semi-analytical method based on the normal approximation of the finite-length coding rate to optimize the allocation of redundancy between index and data bits. To address sequence-level errors, a 5G LDPC code [34] is employed as the inter-oligo code. The main contributions of this paper are summarized as follows.

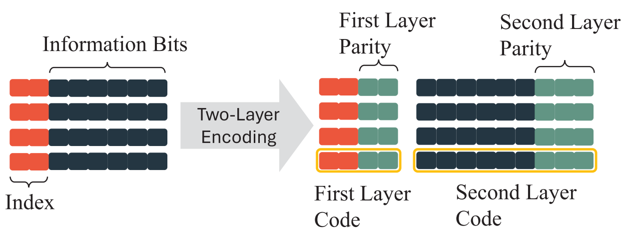
1) Since the importance of the index bits is higher than that of the data bits, we present a novel two-layer intra-oligo coding, in which the index bits are separately protected.
2) Based on the normal approximation of the finite-length coding rates [35], we give a detailed analysis of the frame-error-rate (FER) of the intra-oligo coding when the amounts of redundancy for index bits and data bits are given.
3) We show that we can use the analytical results to optimize the allocation of the redundancies for index bits and data bits in intra-oligo coding when the total amount of redundancy is given. We present extensive analytical and numerical results to show the effectiveness of the proposed analysis in optimizing the redundancy allocation.
4) We conduct system-level simulations, which show that the concatenated coding scheme based on two-layer intra-oligo coding performs significantly better than the concatenated coding scheme based on single-layer intra-oligo coding.

The above results guide the design of reliable DNA-based data storage.

The remainder of the paper is organized as follows. Sec. II gives a brief introduction to the DNA-based data storage system. Sec. III describes the proposed two-layer intra-oligo coding scheme. Sec. IV presents the analysis of the two-layer intra-oligo coding scheme. Sec. V presents extensive analytical and numerical results and Sec. VI concludes the paper.



(a) Single-layer coding scheme



(b) Two-layer coding scheme

Fig. 1. The conventional single-layer coding scheme and the proposed two-layer coding scheme.

II. DNA-Based Data Storage System

A. Overview of the DNA Storage Pipeline

Fig. 2 illustrates various stages of DNA-based data storage.

1) The Encoding Stages: First, the original binary messages are segmented into multiple sequences. This step is necessary because current high-throughput DNA synthesis technologies are limited to synthesizing oligos of approximately 250 bases [27]. Second, the channel encoder introduces redundancy into each sequence to combat the potential data corruption during storage. Finally, the binary-to-DNA transcoder transforms the coded binary data into DNA sequences.

2) The Storage Phase: First, for each DNA sequence, the synthesizer generates a set of DNA oligos, each of which comprises a few hundred bases. Second, DNA oligos may experience degradation, such as hydrolysis and breakage, which are collectively referred to as *decay* [36]. To mitigate DNA loss caused by decay and make sure there are enough copies for reliable information retrieval, PCR can be used to amplify the oligos within the pool. Third, to prepare for sequencing, a portion of the DNA oligos is extracted from the primary oligo pool, referred to as *sampling*.

3) The Retrieval Stage: To retrieve the original messages, the decoding process consists of four steps. First, the sequencer reads the sampled DNA oligos and outputs multiple unordered DNA *reads*. Second, all reads corresponding to the same target sequence are grouped into a cluster. Third, for each cluster, the DNA-to-binary transcoder is then utilized to obtain the binary sequence for decoding. Finally, the FEC decoder is executed to recover the original messages.

B. Error Types in DNA-Based Data Storage System

In the synthesis stage, there exist nucleotide insertions, deletions, and substitutions due to improper activation or nucleotide printing. The error rates in this process range from 0.0005 to 0.005 [37]. Moreover, the synthesis process might be terminated at each loop due to unsuccessful coupling. The probability of successfully coupling one nucleotide to the current strand is called the coupling efficiency, which is about 0.98-0.995 [37].

During storage, DNA oligos decay due to two main reactions: depurination and deamination. Depurination leads to strand breaks and sequence loss. Deamination leads to C-to-U mutations or G-to-T mutations, which may further cause sequence loss [36].

The process of PCR amplification can be employed in synthesis and sequencing. PCR amplification can change the distribution of sequence copies in the branching process of DNA replication. During each PCR cycle, there is a probability P_{pcr} that a sequence will be amplified, referred to as replication efficiency. As PCR is a process with high fidelity (with typical error rates in the order of 10^{-6} to 10^{-5}) [38], the errors introduced within sequences at this stage can generally be neglected.

To retrieve or replicate data, a portion of DNA is extracted from the primary oligo pool. This can be modeled as a random sampling process. This sampling process significantly influences the error characteristics: 1) A sequence will be lost if none of its copies are sampled. 2) The base distribution at a position of a given target sequence may be altered due to sampling, which leads to base errors in the following majority voting. Base substitution is the most common type of error during sequencing, which occurs when the emitted signal from the corresponding base is indistinguishable.

Errors in the DNA storage channel can be further classified into two categories: base-level errors and sequence-level errors [5], [39], [40], [41].

1) Base-Level Errors: In the stages of synthesis, decay, and sequencing, insertion, deletion, and substitution errors will be introduced into the DNA sequence, resulting in base-level errors. Three types of base-level errors are illustrated in Fig. 3.

2) Sequence-Level Errors: In the stage of synthesis, multiple copies corresponding to one target sequence are generated from the oligo pool. However, the numbers of copies of different sequences are unequal due to the possible failure during synthesis. Furthermore, the following stages of decay, PCR, sampling, and sequencing make the numbers of copies of different sequences further uneven. If there are no reads corresponding to a target sequence after sequencing, data from this sequence cannot be recovered. An example of sequence-level errors is given in Fig. 4.

III. The Proposed Two-Layer Coding for Intra-Oligo Coding

A. The Concatenated Coding Scheme

A DNA sequence consists of index bits and data bits. During the majority voting process, base-level errors in the index bits can lead to mis-clustering. Mis-clustered reads act as random noise to the other cluster and alter the proportion of the four base types, disrupting the voting process. In other words, base-level errors in the index bits significantly reduce the precision of the voting process, thereby compromising the fidelity of the recovered message. Hence, adding additional protection to the index bits is important.

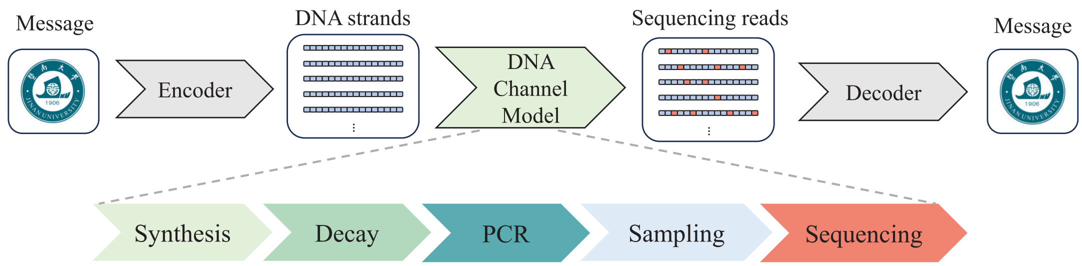


Fig. 2. The processes of the DNA-based data storage.

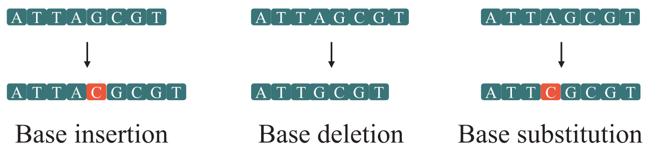


Fig. 3. Examples of base-level errors.

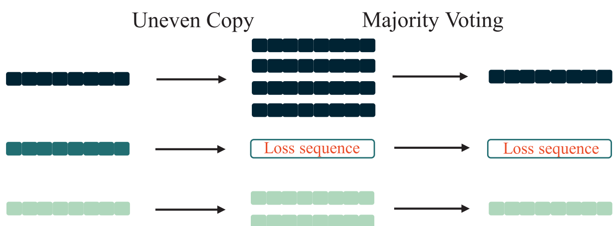


Fig. 4. An example of sequence-level errors.

In this paper, the concatenated coding scheme is employed for DNA-based data storage [42], [43]. Sequence-level errors are corrected with *outer inter-oligo coding* and base-level errors are corrected with *inner intra-oligo coding*. In the following, we propose a novel intra-oligo code referred to as two-layer BCH (TL-BCH) code, which provides additional protection for the index bits and improves the accuracy of the voting result. For inter-oligo coding, we employ the 5G LDPC codes [34]. The coding structure is shown in Fig. 5. We will show that, when compared to the conventional single-layer BCH (SL-BCH) code, this coding scheme significantly reduces the FER of the retrieved message. Details about the proposed two-layer coding scheme are given in Sec. III-B and Sec. III-C.

Some notations used below are defined as follows.

- A finite field of size q is denoted by $\mathbb{F}_q$.
- The set of length-n sequences on $\mathbb{F}_q$ is represented by $\mathbb{F}_q^n$.
- $[n] \triangleq \{1, 2, ..., n\}$ be the set of integers from 1 to n.
- A DNA sequence of length ℓ is denoted by $\mathbf{o}_i \in \{A, C, G, T\}^\ell$, where i is a positive integer.

B. Encoding of the Proposed Two-Layer Coding Scheme

For inter-oligo coding, we employ an (n_0, k_0) LDPC code $\mathcal{C}_0$. For intra-oligo coding, we employ an (n_1, k_1) BCH code $\mathcal{C}_1$ for the index bits and an (n_2, k_2) BCH code $\mathcal{C}_2$ for the data bits. The code rates of $\mathcal{C}_1$ and $\mathcal{C}_2$ are denoted by R_1 and R_2, respectively. The procedure of the concatenated coding scheme based on the proposed two-layer coding is given below.

A binary message $\mathbf{u}$ of length $k_0 k_2$ is segmented into k_0 fragments of length k_2, which is denoted by $\mathbf{u} = (\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_{k_0})$, where $\mathbf{u}_i \in \mathbb{F}_2^{k_2}$, which is a row vector of length k_2. Define the matrices $\mathbf{U}$ and $\mathbf{V}$ as

$$\mathbf{U} = \begin{bmatrix} \mathbf{u}_1 \\ \mathbf{u}_2 \\ \vdots \\ \mathbf{u}_{k_0} \end{bmatrix} \text{ and } \mathbf{V} = \mathbf{U}^T = \begin{bmatrix} \mathbf{v}_1 \\ \mathbf{v}_2 \\ \vdots \\ \mathbf{v}_{k_2} \end{bmatrix}, \tag{1}$$

where $\mathbf{v}_i \in \mathbb{F}_2^{k_0}$, which is a row vector of length k_0. For outer encoding, $\mathbf{v}_i$ is encoded with the encoder of $\mathcal{C}_0$ to obtain $\mathbf{c}_i \in \mathbb{F}_2^{n_0}$, where $i \in [k_2]$. Similarly, we define $\mathbf{C}$ and $\tilde{\mathbf{V}}$ as

$$\mathbf{C} = \begin{bmatrix} \mathbf{c}_1 \\ \mathbf{c}_2 \\ \vdots \\ \mathbf{c}_{k_2} \end{bmatrix} \text{ and } \tilde{\mathbf{V}} = \mathbf{C}^T = \begin{bmatrix} \tilde{\mathbf{v}}_1 \\ \tilde{\mathbf{v}}_2 \\ \vdots \\ \tilde{\mathbf{v}}_{n_0} \end{bmatrix}. \tag{2}$$

To uniquely identify each sequence, an index $\mathbf{z}_i \in \mathbb{F}_2^{k_1}$ is added to each sequence $\tilde{\mathbf{v}}_i$ in the front, where $k_1 \geq \lceil \log(n_0) \rceil$. For the first layer encoding, $\mathbf{z}_i$ is encoded with the encoder of $\mathcal{C}_1$ to obtain the *indexing codeword* $\mathbf{t}_i \in \mathbb{F}_2^{n_1}$. For the second layer encoding, $\tilde{\mathbf{v}}_i$ is encoded with the encoder of $\mathcal{C}_2$ to obtain the *data codeword* $\mathbf{s}_i \in \mathbb{F}_2^{n_2}$. For $\tilde{\mathbf{v}}_i$, the inner encoding output is $\mathbf{w}_i = (\mathbf{t}_i, \mathbf{s}_i) \in \mathbb{F}_2^{n_1+n_2}$.

Finally, for each i, the binary-to-DNA transcoder is applied to $\mathbf{w}_i$, mapping binary codewords to DNA strands $\mathbf{o}_i \in \{A, C, G, T\}^\ell$ following the mapping rule $00 \to A$, $01 \to C$, $10 \to G$, and $11 \to T$, where $\ell = \lceil \frac{n_1+n_2}{2} \rceil$.

Remarks: Note that practical DNA-based data storage systems must satisfy biological constraints that affect synthesis and sequencing reliability. In particular, controlling the GC content and limiting homopolymer runs are essential for reducing errors [5], [18]. To address these constraints, [44] presents a capacity-approaching constrained coding scheme. This scheme is compatible with standard error-correction codes, including the proposed two-layer design. Therefore, it can be incorporated into our framework without altering the underlying channel model or affecting the redundancy optimization methodology. As a result, all analytical conclusions derived in this work remain unchanged.

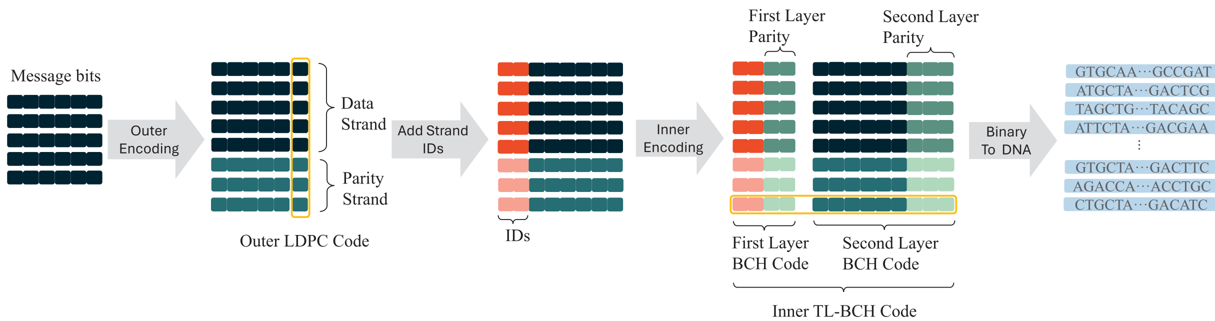


Fig. 5. The proposed concatenated two-layer coding scheme based on outer LDPC codes and inner BCH codes. The outer LDPC codes are encoded in the inter-oligos direction. As a result, each column represents a codeword of the outer LDPC code. The inner BCH codes are encoded in the intra-oligo direction. As a result, each row is the codewords of the inner TL-BCH code. The index bits and the data bits are protected by the first layer and the second layer of the TL-BCH code, respectively. Codewords of the TL-BCH code will be encoded into DNA sequences.

C. Decoding of the Proposed Two-Layer Coding Scheme

The decoding procedure of the concatenated coding scheme based on the TL-BCH code is given below. The outputs of the sequencer in the DNA-based data storage system are θ unordered DNA reads $\hat{\mathbf{o}} = \{\hat{\mathbf{o}}_1, \hat{\mathbf{o}}_2, \ldots, \hat{\mathbf{o}}_\theta\}$, where $\hat{\mathbf{o}}_i \in \{A, C, G, T\}^\ell$. The total number θ of reads is related to the sequencing depth d_{seq} of the sequencer. For each $\hat{\mathbf{o}}_i$, the DNA-to-binary transcoder is employed to obtain the binary sequence $\hat{\mathbf{w}}_i = (\hat{\mathbf{t}}_i, \hat{\mathbf{s}}_i) \in \mathbb{F}_2^{n_1+n_2}$ following the mapping rule $A \to 00$, $C \to 01$, $G \to 10$, and $T \to 11$, where $\hat{\mathbf{t}}_i \in \mathbb{F}_2^{n_1}$ and $\hat{\mathbf{s}}_i \in \mathbb{F}_2^{n_2}$.

For inner decoding, the decoder of $\mathcal{C}_1$ is executed with input $\hat{\mathbf{t}}_i$ to obtain the estimate $\hat{\mathbf{z}}_i$ of the index bits and the decoder of $\mathcal{C}_2$ is executed with input $\hat{\mathbf{s}}_i$ to obtain the estimate $\hat{\mathbf{v}}_i$ of the data bits. Decoded reads with the same index are then clustered, obtaining n_0 clusters of reads $Y_1, Y_2, \ldots, Y_{n_0}$, where Y_i contains the reads with index i and can be empty. Assume that $Y_i = \{\hat{\mathbf{v}}_i^1, \hat{\mathbf{v}}_i^2, \ldots, \hat{\mathbf{v}}_i^{\lambda_i}\}$ are the λ_i reads corresponding to the i-th target sequence. Due to the possible failure of decoding, we have $\sum_{i=1}^{n_0} \lambda_i \leq \theta$. Then, for the j-th bit $v_{i,j}$ of $\mathbf{v}_i$, its probability being equal to 1, referred to as the *voting score*, is estimated as

$$\Pr\{V_{i,j} = 1\} \approx \frac{1}{\lambda_i} \sum_{p=1}^{\lambda_i} v_{i,j}^p \quad \text{for} \quad j = 1, 2, \ldots, n_2, \tag{3}$$

when $\lambda_i > 0$. However, sequence-level errors during storage or the undetected error in the index bits can result in the absence of reads corresponding to a target sequence after sequencing. For this case, we have $\lambda_i = 0$ and we set $\Pr\{V_{i,j} = 1\} = 1/2$.

To initiate the sum-product algorithm (SPA) of the outer LDPC code, we calculate the log-likelihood ratio (LLR) as

$$L_i^j = \begin{cases} -T, & \text{if } \log\left(\frac{1-\Pr\{V_{i,j}=1\}}{\Pr\{V_{i,j}=1\}}\right) \leq -T, \\ T, & \text{if } \log\left(\frac{1-\Pr\{V_{i,j}=1\}}{\Pr\{V_{i,j}=1\}}\right) \geq T, \\ \log\left(\dfrac{1-\Pr\{V_{i,j}=1\}}{\Pr\{V_{i,j}=1\}}\right), & \text{otherwise}, \end{cases} \tag{4}$$

where T is a given threshold.

IV. Redundancy Allocation for Two-Layer BCH Coding Scheme

Redundancy allocation is critical for optimizing the performance of the proposed scheme. It ensures that the limited redundancy budget is optimally allocated between the first layer and the second layer of the proposed two-layer BCH coding scheme. This section studies the impact of redundancies on the reliabilities of the index and data bits, which can be used to optimize the proposed two-layer coding scheme.

A. The Importance of Redundancy Allocation

Let r_1 and r_2 denote the redundancies of $\mathcal{C}_1$ and $\mathcal{C}_2$, respectively. When the redundancy budget $r = r_1 + r_2$ is fixed, decreasing r_1 leads to a higher FER for the index bits and a lower bit-error-rate (BER) for the data bits. A higher FER for the index bits will change the proportion of errors in a specific bit position by introducing incorrectly clustered copies, which will increase the BER of the voting results significantly. In contrast, increasing r_1 leads to a lower FER for index bits and a higher BER for the data bits. To achieve optimal reliability, the impact of redundancy allocation should be analyzed.

Three factors should be considered when studying the error rates of the retrieved data. The first factor is the noise level P_e of the channel. The second factor is the sequencing depth d_{seq} of the sequencer. The third factor is the error-correction capabilities of the selected codes.

B. Impact of r_1 on the FER of the Index Bits

Since the code length of $\mathcal{C}_1$ is n_1, the length of the DNA sub-sequence corresponding to a codeword of $\mathcal{C}_1$ is $\ell_1 = \left\lceil \frac{1}{2} n_1 \right\rceil$. In the DNA storage channel, each base is edited with a probability of $P_s = 0.57 P_e$ [5]. Let X_1 denote the random variable representing the number of edited bases when passing the DNA sub-sequence for indexing through the DNA storage channel. Following [45], we further assume that each base is edited or not independently. Then, the distribution of X_1 is the binomial distribution $X_1 \sim B(\ell_1, P_s)$. That is,

$$\Pr\{X_1 = \xi\} = \binom{\ell_1}{\xi} P_s^\xi (1 - P_s)^{\ell_1 - \xi}. \tag{5}$$

For the symmetric error model, each base has the same probability of being substituted by the other three bases. According to the transcoding rule, substitutions between A and T, C and G lead to 2 bit errors and substitutions between A and C, A and G, T and C, and T and G lead to 1 bit error. When assuming the proportion of each base type is the same, the average number of bit errors is $(2 \times 2 + 1 \times 4)/6 = 4/3$ *bits/edited base*.

We use X_2 to denote the average number of random errors in the retrieved bit sub-sequence corresponding to the indexing codewords. Then,

$$\Pr\left\{X_2 = \frac{4}{3}\xi\right\} = \Pr\{X_1 = \xi\} = \binom{\ell_1}{\xi} P_s^{\xi}(1 - P_s)^{\ell_1 - \xi}. \tag{6}$$

A frame error occurs when the number of bit errors exceeds the error correcting capability t_1 of the BCH code $\mathcal{C}_1$. Hence, the FER ϵ_1 of an index can be approximated as

$$\begin{aligned}\epsilon_1 &= f(r_1, P_e) = \Pr\{X_2 > t_1\} \\ &= \Pr\left\{X_1 > \frac{3}{4}t_1\right\} \\ &= \sum_{i=\lceil \frac{3}{4}t_1 \rceil}^{\ell_1} \binom{\ell_1}{i} (0.57P_e)^i (1 - 0.57P_e)^{\ell_1 - i}.\end{aligned} \tag{7}$$

For a BCH code with redundancy r_1, its correction capability t_1 can be approximated as [46]

$$t_1 \approx \left\lfloor \frac{r_1}{\lceil \log(n_1 + 1) \rceil} \right\rfloor = \left\lfloor \frac{r_1}{\lceil \log(k_1 + r_1 + 1) \rceil} \right\rfloor. \tag{8}$$

Combining Eq. (7) and Eq. (8), the relationship between FER of the indices and r_1 can be obtained. We present a concrete example in Fig. 6 to show the influence of r_1 on the FER with $k_1 = 14$. The analytical results show that, as expected,

- for a given P_e, a higher r_1 corresponds to a lower FER,
- and for a given r_1, a lower P_e leads to a lower FER.

C. Impact of r_2 on the BER of the Data Bits

Similarly, under the assumption of a symmetric error model, each base in the DNA sequence corresponding to the data codeword has a probability of $P_s = 0.57P_e$ being edited. The impact of base-level errors in this situation can be modeled as a binary symmetric channel (BSC) with crossover probability $\delta = \frac{1}{2} \times \frac{4}{3}P_s = 0.38P_e$. Since the code length for data bits ranges from several hundreds to several thousands, we can resort to the normal approximation of the finite-length coding rates [35] to approximate its optimal BER ϵ_2. For the BSC with crossover probability δ, where $\delta \notin \left\{0, \frac{1}{2}, 1\right\}$, the relationship between ϵ_2 and k_2 is given as [35]

$$\begin{aligned}k_2 &= n_2(1 - h(\delta)) \\ &\quad - \sqrt{n_2\delta(1-\delta)}\log\frac{1-\delta}{\delta}Q^{-1}(\epsilon_2) + \frac{1}{2}\log n_2 + O(1),\end{aligned} \tag{9}$$

where $n_2 = k_2 + r_2$ is the code length of the second layer and $Q^{-1}(\cdot)$ denotes the inverse of the Q-function. Rearranging Eq. (9), we obtain

$$\begin{aligned}\epsilon_2 &= g(r_2, P_e) \\ &= Q\left(\frac{n_2(1 - h(\delta)) - k_2 + \frac{1}{2}\log n_2 + O(1)}{\sqrt{n_2\delta(1-\delta)}\log\frac{1-\delta}{\delta}}\right) \\ &= Q\left(\frac{n_2(1 - h(0.38P_e)) - k_2 + \frac{1}{2}\log n_2 + O(1)}{\sqrt{n_2 0.38P_e(1 - 0.38P_e)}\log\frac{1-0.38P_e}{0.38P_e}}\right).\end{aligned} \tag{10}$$

We present a concrete example in Fig. 7 to show the influence of r_2 on the BER for a given $k_2 = 320$ and a given $r = r_1 + r_2 = 112$. The analytical results show that, as expected,

- for a given P_e, a higher r_2 leads to a lower BER,
- and for a given r_2, a lower P_e leads to a lower BER.

D. The Impact of ϵ_1 and ϵ_2 on the FER of Intra-Oligo Coding

In DNA-based data storage, one target sequence will have multiple sequencing reads after the sequencing stage. These sequencing reads will be transcoded into binary sequences. After inner decoding, clustering, and majority voting are performed to obtain a voting score for each bit. These voting scores will be used for outer LDPC decoding. To analyze the impact of ϵ_1 and ϵ_2 on the overall FER after inner decoding and voting, it is essential to study the influence of mis-clustering on the voting scores.

Since sequencing reads with different indices are independent, a frame error of the index will introduce a random noisy sequence to another sequence in the majority voting process. Therefore, the number of sequences within a cluster with correct indices, denoted by λ_c, is determined by the FER of the index bits, and its distribution is given as

$$\Lambda_c \sim B(\lambda, 1 - \epsilon_1), \tag{11}$$

where λ is the number of reads in the cluster. The random number of reads in the cluster is denoted as Λ. Its distribution is derived as

$$\Lambda \sim B\left(\gamma \cdot N_c, \frac{d_{seq}}{n_{syn} \cdot (1 - 0.43P_e) \cdot N_c}\right), \tag{12}$$

where γ denotes the number of remaining copies in the oligo pool for a target sequence, $N_c = \lceil (2P_{pcr})^{N_{pcr}} \rceil$ represents the copy times during PCR amplification, d_{seq} is the sequencing depth, P_{pcr} denotes the replication efficiency of the PCR procedure, n_{syn} denotes the average number of copies for each DNA sequence after synthesis, and N_{pcr} denotes the number of PCR cycles. Due to decaying, the random number Γ of remaining copies for a given sequence in the oligo pool follows the binomial distribution as

$$\Gamma \sim B(n_{syn}, 1 - 0.43P_e). \tag{13}$$

Let's consider a cluster with λ reads. The distribution of Λ_c can be obtained by Eq. (11). Let X_3 be the random number of errors when there are λ_c reads with the correct index. For a given data bit, since the decoded data bits of a read with an incorrect index are uniformly distributed, the distribution of X_3 is then given as

$$X_3 \sim B(\lambda_c, \epsilon_2). \tag{14}$$

For an error to occur, there must be at least $\lambda_c/2$ bit errors in the original λ_c reads. Hence, the probability of error of the consensus sequence, denoted by $\tilde{P}_s$, is given as

$$\tilde{P}_s = \Pr\left\{X_3 > \frac{\lambda_c}{2}\right\} = \sum_{i=\lceil \frac{\lambda_c}{2} \rceil}^{\lfloor \lambda_c \rfloor} \binom{\lceil \lambda_c \rceil}{i} \epsilon_2^i (1-\epsilon_2)^{\lambda_c - i}. \tag{15}$$

Finally, the approximate FER of the consensus sequences after majority voting is given as

$$\epsilon = 1 - (1 - \tilde{P}_s)^{k_2}. \tag{16}$$

Eq. (16) can be used to guide the optimization of the proposed two-layer coding scheme for a fixed overall redundancy r, as shown in the following section.

Remarks: Note that, in the proposed analysis, the error rate of the second layer is approximated with the finite-length bound, and the error rate of the first layer is approximated under bounded distance decoding. The finite-length bound is not used for the first layer due to its lack of accuracy for code lengths less than 100 [35].

V. Numerical Results

This section presents the analytical and simulation results of the proposed two-layer intra-oligo coding scheme. First, we present the analytical FER of the TL-BCH given in Sec. IV to guide the allocation of redundancy. Second, we validate the proposed analytical results by simulating the performance of the two-layer coding scheme using the simulation platform in [37]. Finally, we incorporate the inner two-layer intra-oligo code with optimized redundancy allocation into the concatenated coding scheme, where a 5G LDPC code is taken as the outer code. Similar to [23], [47], and [48], we evaluate the performance of the proposed two-layer coding scheme through simulations, as DNA-based data storage has already been demonstrated experimentally and error correction coding operates purely at the bit level.

A. System Model and Setups

We estimate the error rates of the TL-BCH in a DNA-based data storage system with the DNA storage error simulation pipeline (DESP) [37]. We further modify DESP by including the loss of ordering of the stored sequences to make it more realistic. The input of the DNA storage channel is the n_0 DNA strands generated from the prior encoding process. Let $\mathbf{o} = (\mathbf{o}_1, \mathbf{o}_2, \ldots, \mathbf{o}_{n_0})$ be the input sequences, $\tilde{\mathbf{o}} = (\tilde{\mathbf{o}}_1, \tilde{\mathbf{o}}_2, \ldots, \tilde{\mathbf{o}}_\eta)$ represents synthesized sequences, and $\hat{\mathbf{o}} = (\hat{\mathbf{o}}_1, \hat{\mathbf{o}}_2, \ldots, \hat{\mathbf{o}}_\theta)$ denotes the output of the sequencer, where $\mathbf{o}_i$, $\tilde{\mathbf{o}}_i$, $\hat{\mathbf{o}}_i \in \{A, C, G, T\}^\ell$, $\eta \approx n_0 \times n_{syn}$ and $\theta \approx n_0 \times d_{seq}$. We use P_e to denote the overall error probability. For simplicity, only the dominant reads with the same length are used for decoding [18]. As a result, the loss probability of each sequence $\tilde{\mathbf{o}}_i$ is $P_l = 0.43 \times P_e$, and the substitution probability of each base in $\mathbf{o}_i$ is $P_s = 0.57 \times P_e$ [5].

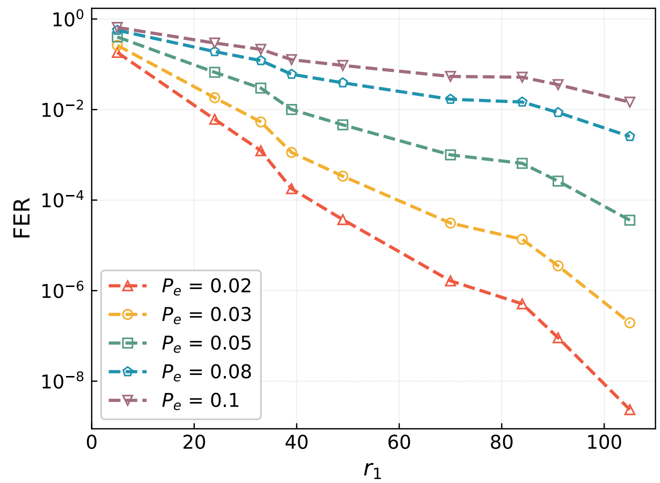


Fig. 6. The impact of the number r_1 of redundancies in the first layer on the FER of the index bits, where we set $k_1 = 14$.

The channel is characterized by the tuple (n_0, ℓ, n_{syn}, d_{seq}, P_{pcr}, C_{pcr}, P_e). The simulations are conducted on synthetic data, where in each run the input sequences are randomly and uniformly generated. The algorithm and DESP are implemented with Python, with a Python-callable interface for encoding/decoding DNA sequences using the MATLAB Communication Toolbox [49].

B. Optimization of the Two-Layer Intra-Oligo Coding

We evaluate the approximate FER when the overall redundancy of the inner two-layer intra-oligo coding is given. We set the data length of the inner code as $k_2 = 320$ bits and the index length as $k_1 = 14$ bits.

Due to the limitations of current DNA synthesis technologies, the maximum length of synthesizable DNA strands remains restricted. For example, Agilent synthesizers can typically produce strands of about 150 to 250 nucleotides [27]. To maintain high storage density, the coding rate of the inner code in DNA-based data storage generally ranges from 0.6 to 0.8 [5], [17], [28]. Accordingly, we set the redundancy per oligo for inner coding, denoted by r, to 92, 112, and 132 bits. These values correspond to inner code lengths of approximately 426, 446, and 466 bits, respectively, when the data length is $k_1 + k_2 = 334$ bits.

The considered redundancy allocations (r_1, r_2) with $r_1 + r_2 = r$ are listed in Table I - III, where we employ BCH codes for $\mathcal{C}_1$ and $\mathcal{C}_2$ for illustration, $R_1 = k_1/n_1$, and $R_2 = k_2/n_2$. The resulting coding scheme is called two-layer BCH (TL-BCH) coding.

In Figs. 8–10, we plot the analytical FERs of the proposed TL-BCH codes using Eq. (16) in Sec. IV-D. For BCH codes, the number of redundant bits cannot be adjusted continuously, making it impossible to match a given target redundancy r exactly. In each simulation, we select the redundancies for the two layers so that their sum is as close as possible to the target r. In the low-error-rate region, we have the following observations.

TABLE I
REDUNDANCY ALLOCATION FOR TL-BCH WITH TOTAL REDUNDANCY ABOUT 92 BITS

Redundancy Allocation	R_1	R_2	Inner Rate	Outer Rate	Overall Rate
(r_1,r_2) = (5, 90)	0.74	0.78	0.75	0.85	0.63
(r_1,r_2) = (10,81)	0.58	0.80	0.75	0.85	0.64
(r_1,r_2) = (15,72)	0.48	0.82	0.76	0.85	0.65
(r_1,r_2) = (24,63)	0.37	0.84	0.76	0.85	0.65
(r_1,r_2) = (47,45)	0.23	0.88	0.75	0.85	0.64

TABLE II
REDUNDANCY ALLOCATION FOR TL-BCH WITH TOTAL REDUNDANCY ABOUT 112 BITS

Redundancy Allocation	R_1	R_2	Inner Rate	Outer Rate	Overall Rate
(r_1,r_2) = (5,108)	0.74	0.75	0.72	0.85	0.61
(r_1,r_2) = (15,99)	0.48	0.76	0.71	0.85	0.61
(r_1,r_2) = (33,81)	0.30	0.80	0.71	0.85	0.61
(r_1,r_2) = (56,54)	0.20	0.86	0.72	0.85	0.61
(r_1,r_2) = (84,27)	0.14	0.92	0.72	0.85	0.61

TABLE III
REDUNDANCY ALLOCATION FOR TL-BCH WITH TOTAL REDUNDANCY ABOUT 132 BITS

Redundancy Allocation	R_1	R_2	Inner Rate	Outer Rate	Overall Rate
(r_1,r_2) = (5, 126)	0.74	0.72	0.69	0.85	0.58
(r_1,r_2) = (15,117)	0.48	0.73	0.69	0.85	0.58
(r_1,r_2) = (24,108)	0.37	0.75	0.69	0.85	0.58
(r_1,r_2) = (45, 90)	0.24	0.78	0.68	0.85	0.58
(r_1,r_2) = (77, 54)	0.15	0.86	0.69	0.85	0.58

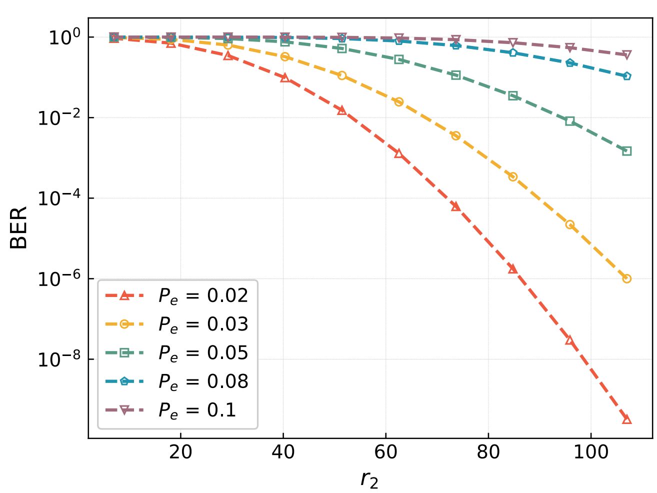


Fig. 7. The impact of the number r_2 of redundancies in the second layer on the BER of the data bits, where we set $k_2 = 320$.

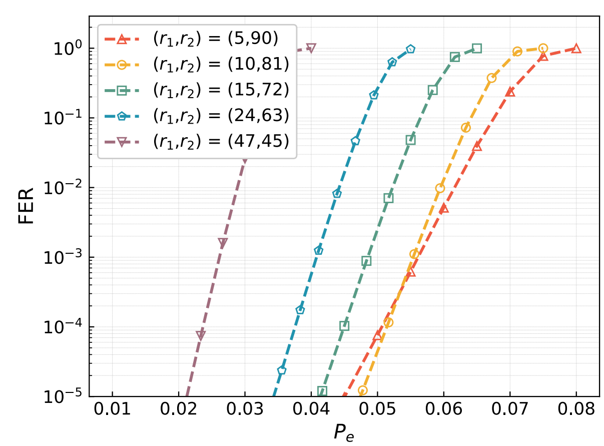


Fig. 8. Analytical FER of the TL-BCH versus P_e. The total redundancies of the inner code are $r \approx 92$ bits, the index length is $k_1 = 14$ bits, the data length is $k_2 = 320$ bits, and the sequencing depth is $d_{seq} = 15$.

1) For $r \approx 92$, the optimal analytical performance is achieved with $(r_1, r_2) = (10, 81)$.
2) For $r \approx 112$, the optimal analytical performance is achieved with $(r_1, r_2) = (15, 99)$.
3) For $r \approx 132$, the optimal analytical performance is achieved with $(r_1, r_2) = (15, 117)$.

Simulated FERs with DESP under different redundancy allocations are shown in Fig. 11, 12 and 13 for $r \approx 92$, $r \approx 112$, and $r \approx 132$, respectively. We have the following observations.

1) For the total redundancy $r \approx 92$, the lowest FER performance in low-error-rate region is achieved with $(r_1, r_2) = (10, 81)$. This is consistent with the analytical results in Fig. 8.
2) For the total redundancy $r \approx 112$, the lowest FER performance in low-error-rate region is achieved with $(r_1, r_2) = (15, 99)$, which is also consistent with the analytical results in Fig. 9.
3) For the total redundancy $r \approx 132$, the lowest FER performance in low-error-rate region is achieved with

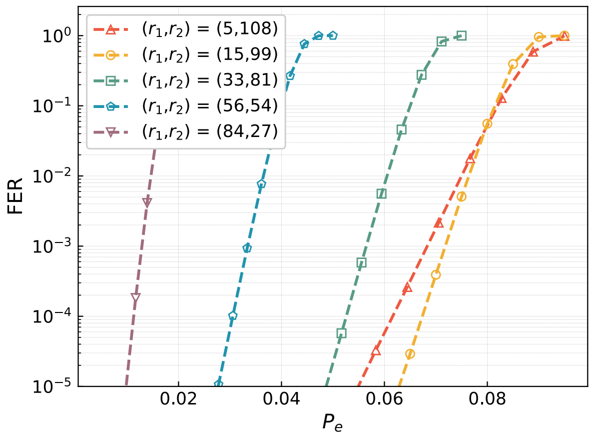


Fig. 9. Analytical FER of the TL-BCH versus P_e. The total redundancies of inner code are $r \approx 112$ bits, the index length is $k_1 = 14$ bits, the data length is $k_2 = 320$ bits, and the sequencing depth is $d_{seq} = 15$.

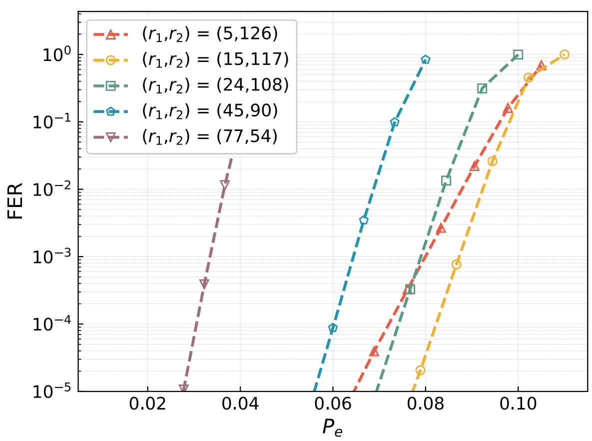


Fig. 10. Analytical FER of the TL-BCH versus P_e. The total redundancies of inner code are $r \approx 132$ bits, the index length is $k_1 = 14$ bits, the data length is $k_2 = 320$ bits, and the sequencing depth is $d_{seq} = 15$.

$(r_1, r_2) = (15, 117)$, which is also consistent with the analytical results in Fig. 10.

The above results indicate the effectiveness of the proposed analytical method in optimizing the proposed two-layer intra-oligo coding scheme. Note that in Figs. 11–13, the trend differs between small and large values of P_e. This behavior can be partially explained as follows. When P_e is large, each cluster contains only a few reads, so the voting process cannot effectively reduce the base-level errors. As a result, the number of residual errors in the first layer exceeds its error-correction capability, and the benefit of the two-layer coding becomes negligible. Conversely, when P_e is small, more reads are available in each cluster, allowing the voting process to better mitigate the base-level errors and thereby enhance the gain of two-layer coding. Given the redundancy budget, the proposed analytical method can provide optimal redundancy allocation for any fixed data and index lengths. This optimization method can also be applied to other coding schemes with a similar indexing strategy, and other error-correcting codes can also be employed in the proposed two-layer coding scheme.

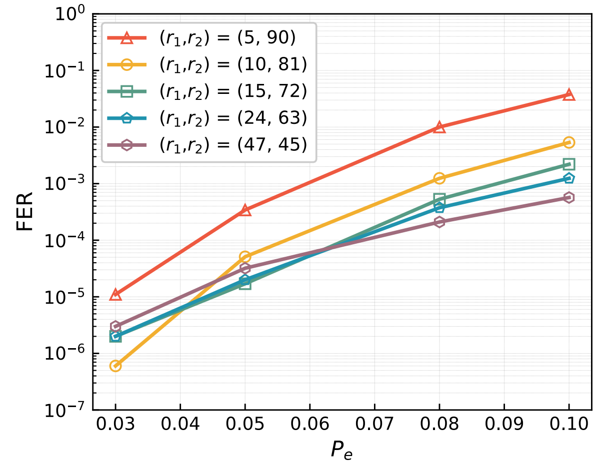


Fig. 11. Simulated FER of the TL-BCH versus P_e. The total redundancy of inner code is $r \approx 92$, the index length is $k_1 = 14$, the data length is $k_2 = 320$, and the sequencing depth is $d_{seq} = 15$.

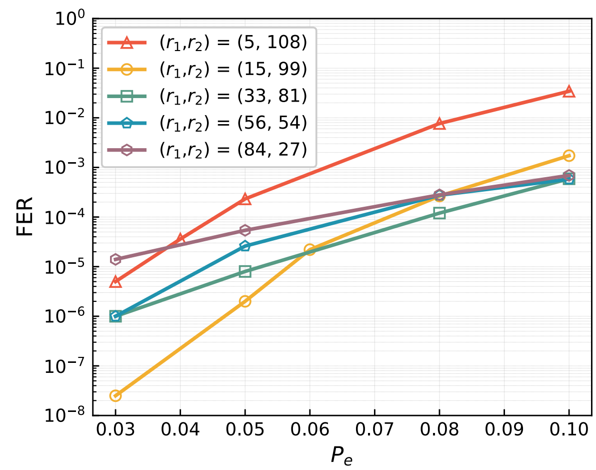


Fig. 12. Simulated FER of the TL-BCH versus P_e. The total redundancy of inner code is $r \approx 112$, the index length is $k_1 = 14$, the data length is $k_2 = 320$, and the sequencing depth is $d_{seq} = 15$.

C. Performances of Concatenated Coding Based on the Proposed Two-Layer Coding

In this subsection, we evaluate the error rates of the concatenated coding scheme built upon the proposed two-layer coding scheme. A 5G LDPC code is employed for outer inter-oligo coding and an optimized TL-BCH code is employed for inner intra-oligo coding. For comparison, we also present the performance of the concatenated coding scheme based on a single-layer inner BCH code. In SL-BCH coding, the index bits and the data bits are combined as a block and then encoded by one BCH code.

We simulate the FERs of the two coding schemes using the error model presented in Sec. V-A, where different channel parameters are considered. The length of the message to be stored is 281.6 KB, and the message is segmented into 7040 sequences of length 320. The encoder of the $(8320, 7040)$ 5G LDPC code generates 8320 outer codewords. The index length is $k_1 = 14$. The overall redundancies for TL-BCH coding and SL-BCH coding are 114 and 117, respectively.

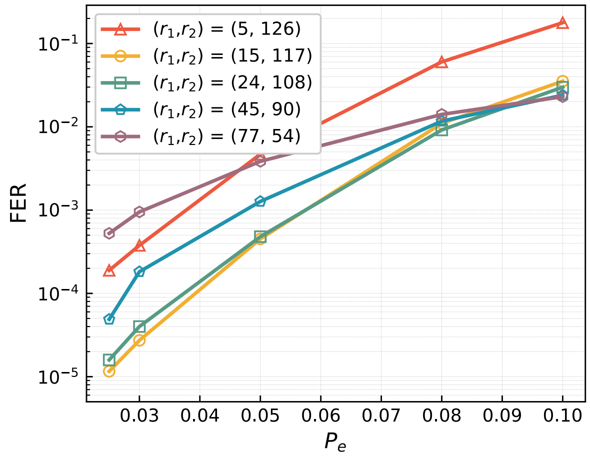


Fig. 13. Simulated FER of the TL-BCH versus P_e. The total redundancy of inner code is $r \approx 132$, the index length is $k_1 = 14$, the data length is $k_2 = 320$, and the sequencing depth is $d_{seq} = 15$.

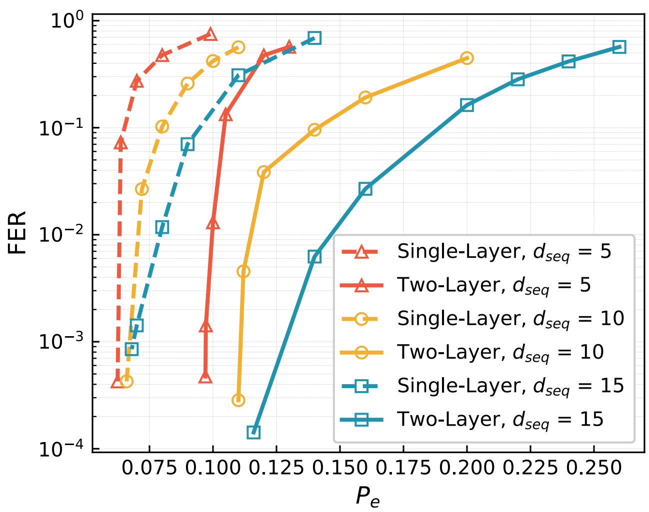


Fig. 14. Simulated FERs of concatenated coding schemes based on two-layer coding and single-layer coding. The index length is $k_1 = 14$ and the data length is $k_2 = 320$. The redundancies for the index bits and the data bits are \text {15} and \text {99}, respectively. We use the (8320, 7040) 5G LDPC code as the outer code.

For the single-layer intra-oligo coding, the $(451, 334)$ shortened BCH code is selected. The inner code rate is $320/451 \approx 0.7095$. Hence, the overall coding rate of the concatenated coding scheme is $0.8462 \times 0.7095 \approx 0.601$. For the two-layer intra-oligo coding scheme, a $(29, 14)$ shortened BCH code is used for the first-layer index coding, and a $(419, 320)$ shortened BCH code is employed for the second-layer data coding, as determined by the optimization results shown in Fig. 9. The rate of inner two-layer coding is $320/(419 + 29) = 0.7143$, and the overall coding rate of the concatenated coding scheme is $0.8462 \times 0.7143 \approx 0.604$. Note that the number of redundant bits of BCH codes cannot be adjusted continuously. This leads to a slight difference in redundancy when comparing the performances. To make the comparison convincing, the coding rates of the simulated TL-BCH are slightly higher than those of the comparable SL-BCH.

The considered editing error rate in the simulation ranges from 0.06 to 0.26. The threshold for LLR is $T = 5$. To show the wide applicability of the proposed two-layer coding scheme, the considered sequencing depths include $d_{seq} = 5, 10, 15$.

The simulation results presented in Fig. 14 show that the error-correcting performances of the concatenated coding scheme based on two-layer coding are significantly better than those of the concatenated coding scheme based on single-layer coding. Compared to the conventional single-layer coding scheme, the proposed two-layer coding scheme enhances reliability in DNA-based data storage without increasing the overall redundancy or sequencing depth. The performance advantage becomes more pronounced at higher sequencing depths, as the two-layer coding scheme limits the error propagation of indexing errors more effectively.

VI. Conclusion

In this work, we proposed a novel two-layer intra-oligo coding scheme tailored to DNA-based data storage. We introduced separate protections for the index bits and data bits, which can combat the base-level errors more efficiently. We then analyzed the impact of redundancy allocation for index and data bits on the overall performance. This analytical result provided an effective approach to optimize the proposed two-layer intra-oligo coding scheme when the redundancy budget is given. Extensive analytical and numerical results validated that the proposed redundancy optimization strategy is effective. Furthermore, simulation results confirmed the advantages of the proposed two-layer coding scheme over the single-layer coding scheme. The proposed two-layer coding scheme and analytical method not only enhance the reliability of DNA-based data storage systems without increasing synthesis or sequencing cost, but also provide design guidance for DNA-based data storage systems that use similar indexing strategies and alternative error-correcting codes. We point out that it is interesting to study the outer LDPC codes, including their code optimization and low-complexity decoding.

Appendix
Outer Code Parameters

The 5G LDPC code used in this work has a block length of $n_0 = 8320$ and a message length of $k_0 = 7040$, yielding a coding rate of $R_0 = k_0/n_0 = 0.846$. The code is constructed based on the 5G NR base graph BG1. For decoding, the maximum number of iterations is set to $I_{\max} = 10$.

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